

Open-AoE: Open-Source Egocentric Data and Toolchain for Embodied AI

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Abstract

Egocentric videos of human manipulation provide scalable supervision for embodied intelligence, yet existing resources rarely combine low-cost continuous capture, manipulation-level structured annotations, and reusable tools for robot learning. We present **Open-AoE**, an open, community-driven egocentric manipulation dataset and toolchain spanning the full pipeline from smartphone capture to model training. Its first release contains approximately 2,000 hours of manipulation video collected in natural environments by more than 1,000 participants using over 400 smartphones. The dataset provides text annotations, MANO-based hand poses, camera trajectories, and temporally localized atomic actions. Open-AoE further includes a data processing pipeline that transforms raw recordings into structured samples through temporal action segmentation, semantic annotation, and hand and camera reconstruction. A separate downstream toolchain supports visualization, cross-embodiment retargeting, model-specific data conversion, and training recipes for VLA, VAM/WAM, and world models. By integrating scalable capture, structured processing, and downstream adaptation, Open-AoE reduces the barriers to both data contribution and reuse, providing practical open infrastructure for embodied model training, human-to-robot transfer, and world modeling.

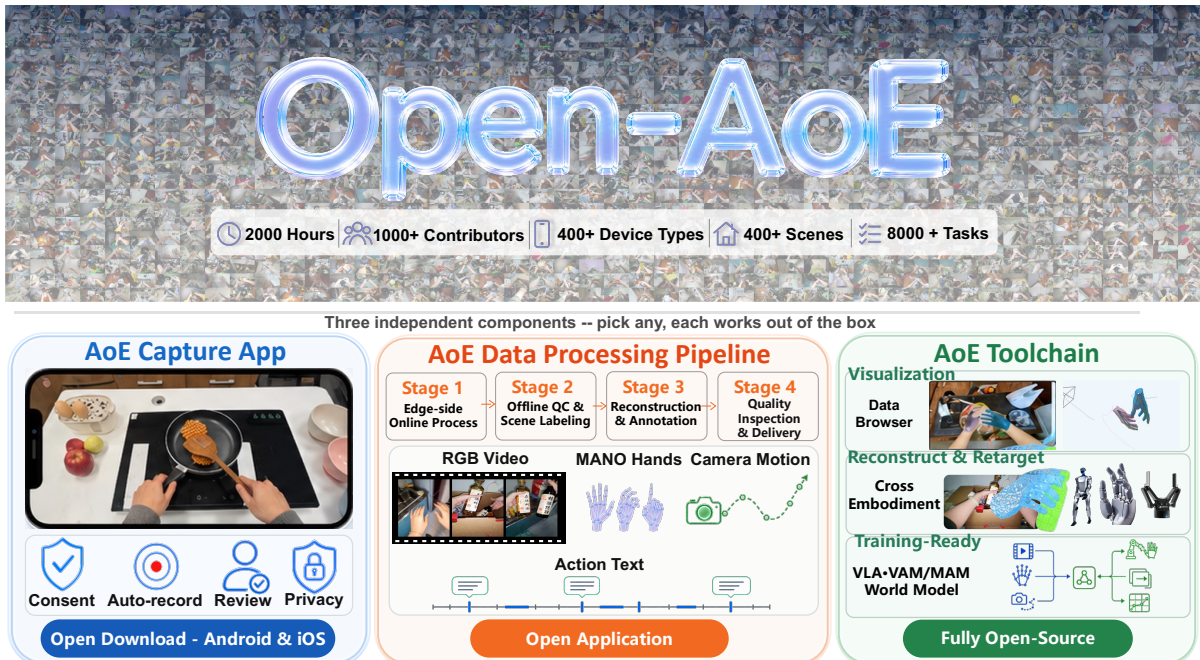


Figure 1: **Overview of Open-AoE.** Open-AoE provides independent modules for smartphone capture, structured data processing, and downstream robot learning, supporting 2,000 hours of data across 1,000+ contributors, 400+ device types, 400+ scenes, and 8,000+ tasks.

Table 1: Comparison of egocentric manipulation datasets. Device diversity counts distinct capture device models. Open-AoE combines large-scale consumer-smartphone collection with complete manipulation annotations and a data-to-model toolchain.

Dataset	Hours	Tasks	Hand Pose	Language	Camera Traj.	Collectors	Device Diversity	Retarget	Training
Ego4D [1]	3,670	N/A	✗	✓	✗	931	2 models	✗	✗
EPIC-KITCHENS [2]	100	N/A	✗	✓	✗	37	2 models	✗	✗
EgoDex [3]	829	194	✓ _{native}	✓	✓	N/A	1 model	✗	✗
EgoLive [4]	1,680	346	✓	✓	✓	N/A	1 custom model	✗	✗
OpenEgo [5]	1,107	290	✓ _{unified}	✓	Partial	6 sources	Mixed models	✗	✗
EgoScale [6]	20,854	1,000+	✓	✓	✗	N/A	1 model	✗	✗
Open-AoE	2,000	8,000+	✓ _{MANO}	✓	✓	1,000+	400+ models	✓	✓

1. Introduction

Embodied foundation models are increasingly constrained by the scale and quality of real-world interaction data [7, 8]. Unlike language models, which can learn from internet-scale text, robot learning requires physical demonstrations that capture how humans perceive scenes, move their hands, contact objects, and complete temporally extended tasks [8]. Such data must be not only large and diverse, but also structured in geometry, time, and semantics for manipulation learning [8]. Empirical scaling studies further show that diversity across environments and objects can matter more than simply increasing the number of demonstrations within fixed conditions [7].

Egocentric video provides a natural and scalable modality for this problem [1, 2]. By recording interaction from the actor’s viewpoint, it captures hands, objects, scenes, and action progress in the same visual stream [1, 2]. Compared with third-person video, egocentric video is closer to the perceptual structure of robot execution [3, 6]. Compared with robot teleoperation, it can be collected more naturally, continuously, and cheaply in everyday environments [9, 8]. Recent datasets such as Ego4D, EPIC-KITCHENS, EgoDex, EgoLive, OpenEgo, and EgoScale have advanced egocentric data along multiple axes, including larger-scale daily activity capture, finer-grained hand pose annotation, richer manipulation scenarios, and stronger alignment with robot learning [1, 2, 3, 4, 5, 6].

However, a central gap remains. The community does not simply need more egocentric video. It needs open data infrastructure that can continuously collect, systematically process, and directly support model training. High-quality egocentric datasets often rely on specialized head-mounted devices, AR/VR hardware, or controlled capture protocols [3, 4]. Specialized hardware can improve pose and camera sensing, but it narrows accessibility relative to commodity-smartphone collection [3, 4, 9]. In contrast, large-scale passive video is easier to obtain, but usually lacks the hand motion, camera trajectory, action boundaries, and physical structure required for robot training [3, 5]. Recent efforts consequently introduce dataset-specific pipelines for hand-pose unification, human-to-robot transfer, or raw-video processing [5, 6, 9]. Releasing videos is therefore not the same as releasing trainable data.

To address this gap, we introduce **Open-AoE**, a community-oriented egocentric manipulation dataset and toolchain. Open-AoE builds on the AoE consumer-smartphone collection framework [9] and releases the full pipeline from raw capture to model training. The first release contains approximately **2,000 hours** of egocentric human manipulation data collected in natural environments by **1,000+ participants** using **400+ smartphone models**. The dataset provides structured signals including videos, English language descriptions, MANO hand poses, camera trajectories, and atomic action segments. For dataset production, Open-AoE releases the

cloud-side processing pipeline that converts uploaded smartphone clips into structured samples through quality screening and privacy erasure, temporal segmentation, semantic labeling, camera-trajectory estimation, hand reconstruction, atomic-action annotation, and multi-stage quality inspection. Separately, for data consumption, the downstream toolchain starts from the released corpus and supports synchronized visual inspection, 4D hand-object interaction reconstruction, cross-embodiment motion retargeting, and robotized-video generation. AoE-Training-Ready then maps the aligned video, hand, camera, and language signals into action representations and training interfaces for three model directions: Vision-Language-Action (VLA) policies, World Action Models (WAMs), and World Models.

The goal of Open-AoE is not to release an isolated video collection, but to open a reusable **capture-process-reconstruct-train** data production loop. Starting from the same egocentric corpus, researchers can inspect data, reconstruct 4D interactions, replay motions on robots, train vision-language-action models, learn world action models, or study world model learning without rebuilding a private post-processing stack. By lowering both the barrier to contributing data and the barrier to using data, Open-AoE aims to turn low-cost smartphone-captured human manipulation video into practical open infrastructure for embodied intelligence research.

Our contributions are three-fold:

- We release approximately **2,000 hours** of real egocentric human manipulation data collected with consumer-grade smartphones in natural environments, covering diverse participants, devices, and everyday manipulation scenarios.
- We open an end-to-end processing pipeline that converts raw smartphone videos into structured training samples with hand poses, camera trajectories, action segments, and language descriptions.
- We provide a downstream embodied learning toolchain for data visualization, reconstruction, and cross-embodiment retargeting, together with training-ready representations and interfaces for three major model directions: Vision-Language-Action (VLA) policies, World Action Models (WAMs), and World Models.

2. Related Work

2.1. Open Egocentric Data Sources for Embodied Learning

Open egocentric corpora differ chiefly in the supervision they expose above raw video. EPIC-KITCHENS-100 [2] and Ego4D [1] provide broad coverage of daily activities, environments, and language. Geometry-rich releases make manipulation more directly observable: HOI4D [10] provides RGB-D sequences with hand and object poses, segmentation, and reconstructed geometry; HOT3D [11] adds motion-capture ground truth for multi-view hand, object, and camera poses; EgoMimic [12] pairs camera motion and 3D hand tracking for imitation learning; and EgoDex [3] scales native hand tracking and language annotation to 829 hours of dexterous tabletop manipulation.

Other releases address normalization, scale, or sensor fidelity. OpenEgo [5] consolidates 1,107 hours from six public datasets into standardized hand poses and localized action primitives. EgoScale [6] studies dexterous policy scaling with more than 20,000 hours of in-the-wild video, while EgoLive [4] contributes 1,680 hours of 60-FPS stereo recordings from service scenarios with camera motion, 3D hands, depth, masks, and sub-task descriptions. Open-AoE [9] complements these efforts with a 2,000-hour smartphone-captured release spanning 1,000+ contributors, 400+ device types, 400+ scenes, and 8,000+ tasks, together with atomic action descriptions, MANO

hand motion, camera trajectories, and reusable processing and model-integration tools.

2.2. Embodied Applications of Egocentric Data

Reconstruction and cross-embodiment transfer. Egocentric video becomes actionable when implicit motion is converted into geometric or control-oriented supervision. HaWoR [13] recovers world-space hand motion from monocular video, while EgoInfinity [14] reconstructs metric 4D hand-object trajectories and supports robot retargeting. EgoAERO [15] estimates contact-consistent trajectories from a single RGB-D demonstration, and EgoEngine [16] jointly generates robotized observations and feasible robot actions. These outputs support several downstream uses: reconstructed hand-object assets expose contact and object motion, retargeted trajectories provide robot-facing supervision, and robotized videos preserve scene context while narrowing the visual embodiment gap. Visual adaptation, shared action spaces, active vision, and whole-body retargeting further address viewpoint and morphology differences across embodiments [17, 18, 19, 20, 21].

Vision-Language-Action policies. VLA methods learn action-aware representations directly from human video rather than requiring every example to be replayed by a robot. Being-H0 [22] tokenizes human motion for vision-language-action pretraining; EgoVLA [23] predicts wrist and hand actions before robot adaptation; and VITRA [24] derives robot-aligned supervision from real-life activities. EgoScale [6] further shows benefits from scaling diverse egocentric pretraining. The route from human observation to robot policy can therefore combine language-aligned task semantics with embodiment-aware action targets, but it depends on temporally stable segments, geometry, and action interfaces that remain meaningful after embodiment conversion.

World Action Models. WAMs occupy a middle ground between direct policy learning and unconstrained video generation: they learn action variables together with observation dynamics, including settings where explicit robot actions are unavailable. Latent-action learning can infer compact motion surrogates from video, although visual distractors may require additional supervision [25]. LaWAM [26] uses compact latent visual subgoals to reduce pixel-space generation latency, while DreamZero [27] jointly predicts future video and action and uses the learned dynamics as a zero-shot policy. This route makes temporal alignment between the current observation, the inferred or supplied action, and the resulting visual change especially important.

World Models. Predictive world models use egocentric video to learn reusable scene and interaction dynamics beyond a single robot policy. DreamDojo [28] pretrains a generalist robot world model on 44,000 hours of egocentric video with continuous latent actions; iVideoGPT [29] studies scalable interactive video prediction; and Ctrl-World [30] introduces controllable generation for robot manipulation. Such models can exploit both weak latent actions and explicit hand, camera, or robot controls, provided that long videos are segmented into coherent transitions and paired with stable conditioning signals. Across reconstruction, retargeting, VLA, WAM, and world-model learning, the shared requirement is an interface between diverse human video and model-consumable supervision. Open-AoE is designed to provide both sides of that interface through a broad smartphone corpus and its associated processing and conversion tools.

3. Open-AoE Dataset

3.1. Data Processing Pipeline

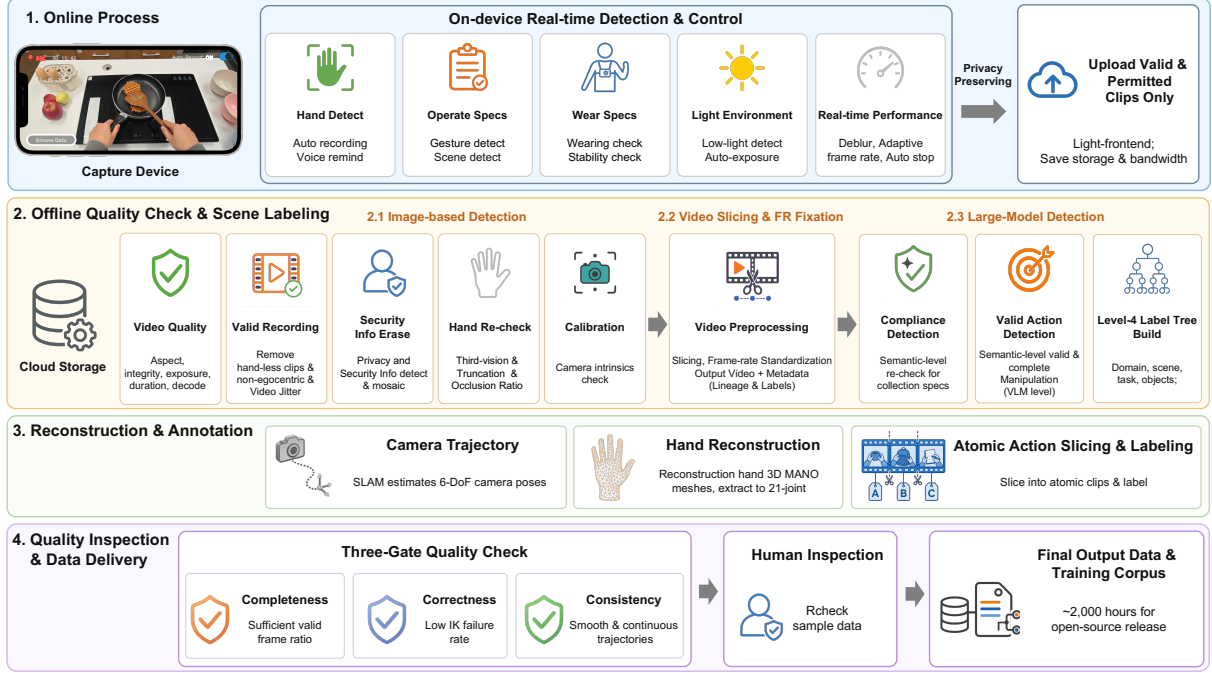


Figure 2: **Open-AoE data processing pipeline.** Starting from edge-side on-device detection and control, raw egocentric videos pass through offline quality checking and coarse labeling (image-based detection, video slicing, and large-model detection), followed by pose reconstruction, atomic-action annotation, and three-gate quality inspection to produce a privacy-filtered training corpus. The released subset contains approximately 2,000 hours of egocentric manipulation video captured with consumer smartphones, together with English annotations, MANO hand poses, and camera trajectories.

The pipeline covers both the online capture side and the offline processing side shown in the flowchart. It proceeds through four stages, namely edge-side online detection, processing and pre-analysis, reconstruction and annotation, and quality inspection with data delivery, and ends by producing the desensitized training corpus.

3.2. Edge-side Online Detection

On the capture device, lightweight on-device vision models gate recording in real time and govern capture quality, so that only clips containing meaningful hand-object interaction are retained [9]. A hand-visibility check starts recording automatically once hands appear and stops it when they leave the frame, and prompts the user by voice to readjust whenever a hand reaches the image border or becomes truncated, which keeps the manipulating subject fully visible. Operating and wearing compliance are monitored at the same time, with non-standard gestures and improper wearing poses corrected through real-time voice feedback. The device also adapts to the lighting environment by turning on a constant fill light together with global auto-exposure under dim light, and it sustains real-time performance through a fixed focal length, motion stabilization, deblurring, and an adaptive frame rate, halting recording when storage runs low or the phone overheats. This light-frontend design protects privacy at the

source, saves storage and bandwidth, and uploads only the relevant clips to the cloud for offline processing.

3.3. Processing & Pre-Analysis: Offline Quality Check and Gross Labeling

3.3.1. Image-based Detection

The first gate uses no deep models and works from sampled frames with image analysis and lightweight detectors. For video quality, a bank of rule-based detectors runs in parallel to flag the common defects that make footage unusable, covering aspect ratio and rotation, file-header and stream integrity, pure-black or over- and under-exposed frames, insufficient duration, frame-rate instability and frame drops, and decoding integrity, so that clearly corrupted videos are discarded early. A valid-action check then removes meaningless segments and non-egocentric footage such as fixed-mount or third-person side views, leaving only valid first-person manipulation. Privacy erasure detects faces and other sensitive content and replaces the personally identifiable information in the metadata, including the collector’s name and contact, with anonymized identifiers while rewriting the corresponding fields and directory names; the mapping is stored in full and supports incremental reuse, so the process can be safely re-run and the anonymization reversed when authorized. A hand re-check verifies that both hands are visible, and to avoid reacting to a single noisy frame it confirms a defect only when it persists across several consecutive frames. Calibration finally validates the camera intrinsics for completeness and reasonableness, since reliable calibration is a prerequisite for later pose reconstruction.

3.3.2. Video Slicing and Frame-Rate Fixation

After image-based detection, the video is split along contiguous qualified and unqualified spans and its frame rate is fixed, so that a long recording becomes several semantically independent Parts, with qualified segments that are too short flagged separately. Each Part stays linked to its source video through a unified process-tag file that records the data lineage from the raw video down to the Part and accumulates multi-level labels as the pipeline advances, giving a data contract that holds across the whole process.

3.3.3. Large-Model Detection

Qualified Parts then move to the large-model stage, where the vision-language model Qwen3.7-Plus adds higher-level semantic detection and gross labeling. Compliance detection re-checks at the semantic level whether a segment follows the collection specifications and catches violations that image-based detection can hardly see, while valid-action detection judges whether the action is valid and forms a complete, purposeful manipulation, complementing the earlier image-level screening. The same model then carries out a four-level label-tree build on a closed-set design, assigning a representative frame exactly one scene and one task from predefined sets together with the salient objects it contains, after which these outputs are standardized into unified identifiers and accumulated over the user, scene, task, and object dimensions to form a hierarchical label system.

3.4. Reconstruction & Annotation

Each qualified Part enters reconstruction, where camera-trajectory estimation, hand reconstruction, and atomic-action annotation are produced together. Camera poses are recovered

by DROID-W [31], whose robust kernels are re-tuned for handheld and wearable capture to keep the 6-DoF trajectory stable under strong shaking and moving-torso interference. Hand reconstruction starts from a two-hand detector trained on large-scale egocentric data collected through AoE, which supplies per-frame bounding boxes, links detections across frames, and fills in missing or occluded hands; building on these boxes, HaWoR [13] recovers 3D MANO [32] meshes that are metric-scale aligned through SLAM and brought into world coordinates, with sliding-window optimization enforcing temporal consistency. A global bundle adjustment over the HaWoR reconstruction and the dynamic-object masks from DROID-W then optimizes the hand meshes and the camera trajectory together in one world frame, removing the scale drift and coordinate mismatch that arise across segment-wise reconstructions. From the recovered meshes, 21-joint hand keypoints are read out, and the video is finally cut into semantically coherent atomic clips labeled in English, with a human-in-the-loop review correcting model hallucinations.

3.5. Quality Inspection and Data Delivery

The reconstructed and annotated Parts pass a three-gate quality check before archiving, from which about 2,000 hours are curated out of the full pool for open-source release. The completeness gate keeps Parts whose hand-reconstruction valid-frame ratio is high enough for reliable pose annotation; the correctness gate keeps those whose inverse-kinematics failure rate stays low when retargeting to the 28-DoF joint space, which preserves downstream usability; and the consistency gate favors Parts with smooth, continuous camera trajectories free of abrupt jumps between adjacent frames. A small random sample is finally inspected by humans to confirm annotation accuracy and to drop edge cases such as severe occlusion or extreme lighting, yielding the desensitized training corpus that constitutes the final output.

3.6. Data Distribution Analysis

Figure 3 summarizes Open-AoE diversity at two complementary scopes: the semantic, context, and contributor panels describe the audited 100-hour delivery, whereas the device-model and horizontal-field-of-view panels use the full consumer-phone registry. Wedge area, bar height, and word size encode relative prevalence or frequency rather than absolute duration.

Broad scenes, actions, and interaction objects. The 100-hour audit covers **46,802 atomic-action intervals**, **193 verb strings**, **7,346 object phrases**, and 522 raw context descriptions. These distributions cover reusable manipulation primitives together with a long tail of foods, tools, containers, craft materials, garments, and household objects. Collection extends beyond kitchens and tabletops into offices, workshops, bedrooms, bathrooms, outdoor areas, laundry spaces, and retail or service settings. The joint diversity of scene, action, and interaction object provides a substrate for compositional generalization.

A broad consumer-camera domain. The full registry resolves **889 device-model IDs** to **818 market models**, of which 876 IDs have calibrated horizontal FOV. To our knowledge, this is the broadest reported consumer-phone device coverage for an egocentric manipulation dataset. Device identity is also a practical proxy for the underlying camera stack, including the lens, sensor, image-signal processor, resolution mode, exposure pipeline, and capture firmware. Compared with a single-headset collection, Open-AoE therefore introduces naturally occurring camera-domain variation at capture time.

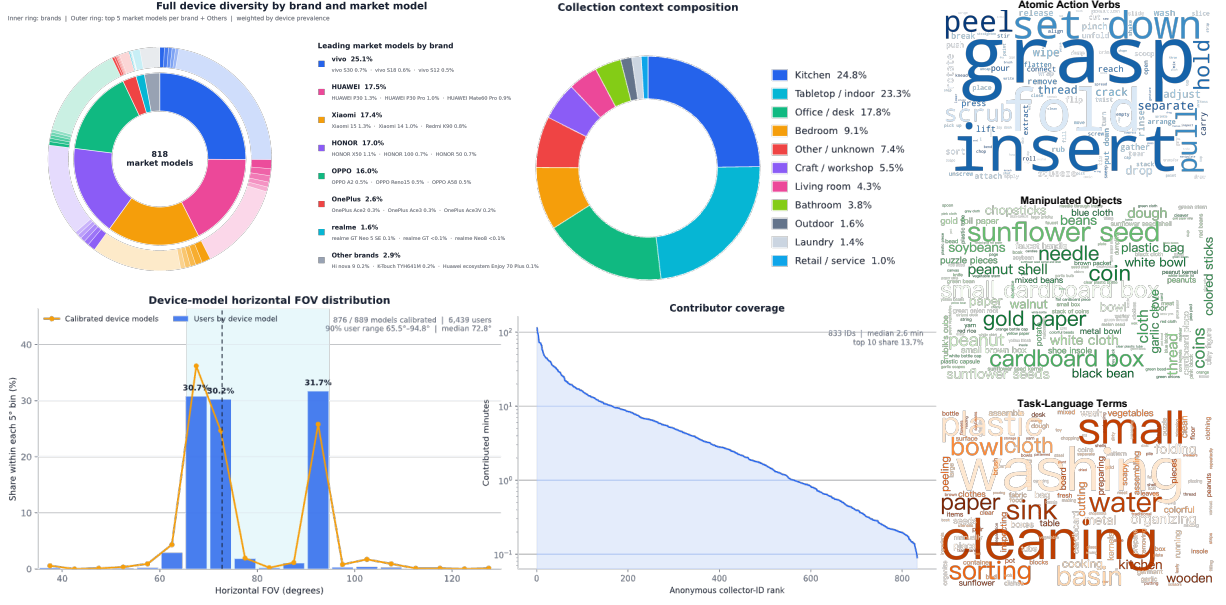


Figure 3: **Open-AoE collection, sensor, and semantic diversity.** The nested donut summarizes brand and market-model coverage in the full device registry, and the lower-left panel shows the device-model-weighted horizontal-FOV distribution. The center panels report context composition and anonymous contributor coverage in the 100-hour audit, while the word clouds summarize atomic-action verbs, manipulated objects, and task-language terms. Device and FOV statistics use the full phone registry; all other panels use the audited 100-hour delivery.

Optical diversity for robust visual transfer. Calibrated horizontal FOV spans 35.1° to 125.7° , with the central 90% of the device-model-weighted distribution between 65.5° and 94.8° . The multimodal profile reflects recurring camera and lens configurations rather than one optical system. A downstream visual policy must recover hand-object geometry and motion across different projections, crops, distortions, and imaging pipelines. This variation provides a form of natural sensor-domain randomization that may improve robustness when transferring to unseen cameras, deployment rigs, and robot viewpoints.

Takeaway. Open-AoE is differentiated not only by scale, but by diversity at three coupled levels: what people manipulate, where and how they act, and the consumer-camera domain through which every interaction is observed. Together, these properties create a stronger basis for visually robust, cross-domain manipulation learning.

Scope and claim boundary. Semantic, context, and contributor counts describe the audited 100-hour delivery; device and FOV coverage use the full registry. The downstream benefit of camera-domain diversity is a training hypothesis that should be validated through controlled ablations.

4. Open-AoE Toolchain

The Open-AoE Toolchain converts egocentric manipulation data into research assets that can be inspected, reconstructed, retargeted, and used for training. Each AoE sample synchronizes undistorted RGB video, camera parameters and trajectories, MANO hand reconstructions [32], validity masks, and atomic action annotations. These signals jointly describe visual context, hand motion, camera ego-motion, and task semantics, yet downstream methods consume different

interfaces. The toolchain therefore projects synchronized data into task-specific representation spaces rather than enforcing a universal format.

4.1. AoE-Visualization

AoE-Visualization is the entry point for synchronized data browsing. It uses the undistorted egocentric video as the canvas and overlays MANO hand meshes, 21-keypoint hand skeletons, future wrist trajectories, atomic-action annotations, a world-frame 3D view, and a timeline. A single review video makes image evidence, hand geometry, camera motion, and semantic labels directly observable as temporally aligned signals. It also provides a unified visual format for dataset examples in the paper.

4.2. Reconstruction and Retargeting

Reconstruction and Retargeting connects AoE egocentric video to three robotized outputs, as shown in Figure 4.

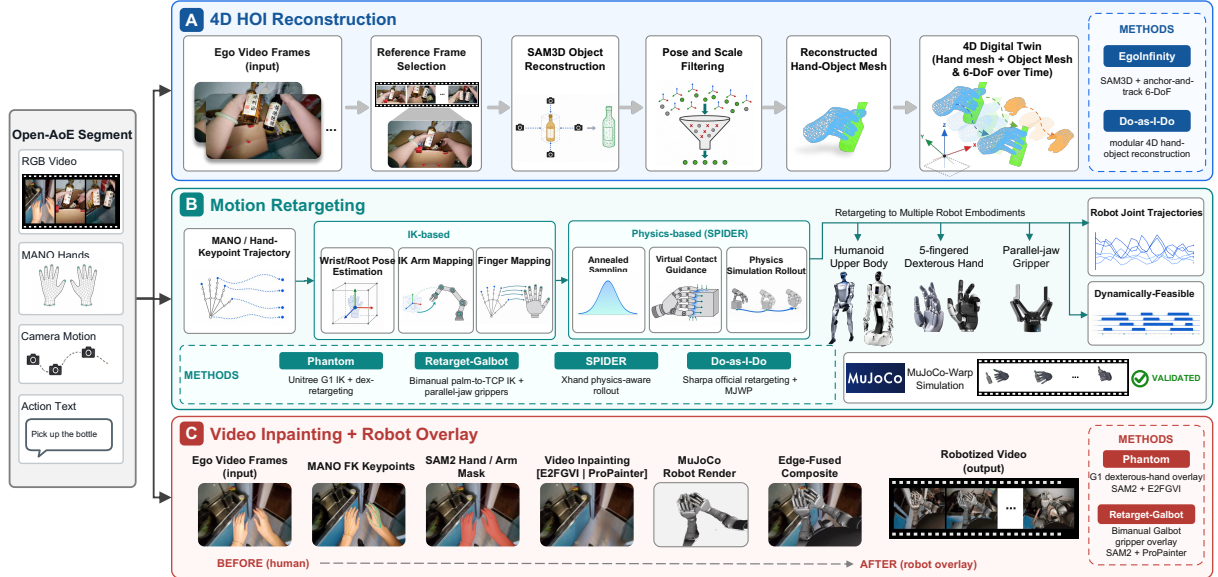


Figure 4: Reconstruction and Retargeting. A smartphone-captured AoE egocentric video can be projected into three complementary outputs: 4D hand-object interaction assets, robot-usable motion, and robotized video. The toolchain integrates EgoInfinity and Do-as-I-Do for 4D reconstruction; Phantom, Retarget-Galbot, SPIDER/XHand, and Do-as-I-Do/Sharpa for motion retargeting; and Phantom or Retarget-Galbot for SAM2-based video inpainting and MuJoCo robot overlay.

The first route is 4D hand-object interaction reconstruction, which recovers hand meshes, object 6-DoF trajectories, contact relations, and textureable assets to build an analyzable 4D digital twin. Open-AoE supports this route with camera trajectories, MANO hand priors, and action-segment semantics, so keyframe selection, object tracking, scale diagnosis, and contact analysis need not rely only on image appearance. The toolchain integrates EgoInfinity [14] and Do-as-I-Do [33] to lift egocentric video into hand-object assets with geometric and contact semantics.

The second route is motion retargeting, which converts AoE MANO or keypoint trajectories, or hand and object trajectories produced by 4D reconstruction, into robot-usable trajectories

or control sequences. Phantom maps AoE hand motion to Unitree G1 with Dex3 or Inspire hands, while Retarget-Galbot maps bimanual palm targets to the Galbot/Galaxea arms and parallel-jaw grippers. Retarget Lab further integrates EgoInfinity/G1, Do-as-I-Do/Sharpa, and SPIDER/XHand, covering both kinematic and physics-aware routes for cross-embodiment retargeting [34, 35]. These methods map human wrist, finger, and object motion to humanoid robots, dexterous hands, or parallel-jaw grippers, and use MuJoCo simulation to produce robot joint trajectories and executable motion.

The third route is video inpainting plus robot overlay. It removes the human arm from a real egocentric video and composites MuJoCo-rendered robot execution back into the original scene, producing robotized video. Phantom combines MANO forward-kinematics keypoints, SAM2, E2FGVI, and G1 rendering, while Retarget-Galbot combines SAM2, ProPainter, and bimanual Galbot gripper rendering. The real environment, manipulated object, and task progression are preserved while the actor appearance is replaced by a robot, enabling visualization of retargeting results and analysis of robot-scene consistency. Together, the three routes answer what happened, how a robot can execute it, and how robot execution looks in the original scene.

4.3. AoE-Training-Ready

AoE-Training-Ready adapts structured AoE data to model-training interfaces, treating conversion as action-semantics selection rather than file-format translation.

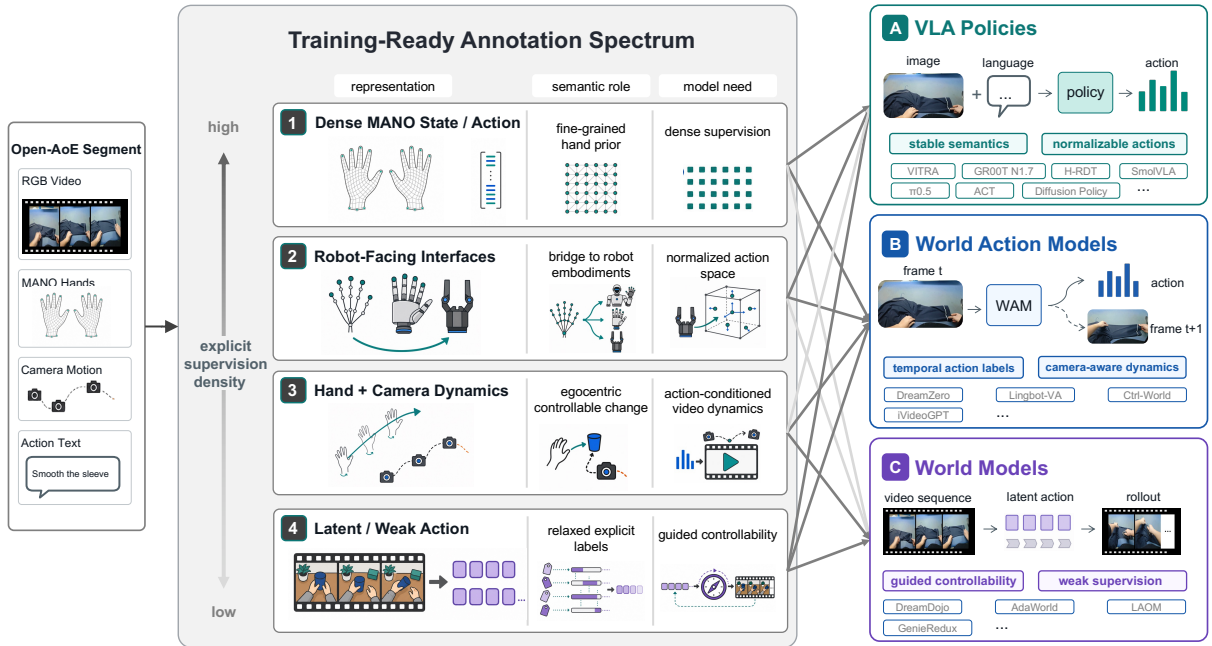


Figure 5: AoE-Training-Ready. A synchronized AoE segment can be converted into action representations with different supervision densities and semantic roles. The integrated training recipes connect dense MANO, robot-facing, hand-plus-camera, and latent or weak action representations to Vision-Language-Action policies, World Action Models, and World Models.

For dense supervision, the repository defines a shared 110D bimanual state/action vector. Each hand contains validity, wrist translation, wrist axis-angle rotation, wrist velocity, and a 45D MANO finger pose; state uses the current frame, while action uses the next-frame target at the method-specific stride. Robot-facing interfaces additionally include the 48D bimanual wrist-and-fingertip action used by H-RDT and the 62D Sharpa or 20D gripper representations

used by GR00T N1.7. A second shared representation for egocentric models uses a separate 22D bimanual state and 20D hand action, with optional 6D camera ego-motion appended to form a 26D hand-plus-camera action.

The VLA recipes integrate VITRA [24], GR00T N1.7 [36], H-RDT [37], SmolVLA [38], and LeRobot-based [39] $\pi_{0.5}$, ACT, and Diffusion Policy training. These recipes adapt the synchronized AoE signals to dense MANO, wrist-and-fingertip, Sharpa, gripper, or hand-plus-camera action spaces according to model requirements.

The World Action Model and video-action recipes integrate DreamZero [27], LingBot-VA [40], Ctrl-World [30], and iVideoGPT [29]. DreamZero, LingBot-VA, and Ctrl-World use temporally aligned current-hand states and next-frame action targets, while iVideoGPT uses 20D hand or 26D hand-plus-camera actions to condition video dynamics. This design covers both action prediction and action-conditioned visual generation while explicitly addressing the strong dynamics caused by egocentric camera motion.

The World Model recipes integrate GenieRedux [41], LAOM [25], AdaWorld, and Dream-Dojo [28], covering unsupervised latent actions, limited hand supervision, true-action guidance, and MANO-conditioned training. Thus, the contribution of Training-Ready is not a set of isolated converters, but a view of how one AoE annotation source can serve different action semantics and model families. Together, Visualization, Reconstruction and Retargeting, and Training-Ready provide a reproducible engineering path from consumer-smartphone capture to robot replay, VLA training, video-action modeling, and world-model learning.

5. Dataset Analysis

To assess whether Open-AoE exhibits a distinctive data profile beyond dataset scale, we compare it with OpenEgo, EgoDex, and EgoXtreme across four complementary dimensions: visual representation diversity, semantic and temporal supervision, annotation consistency, and downstream training utility. Approximately 100 hours of data are included for Open-AoE, OpenEgo, and EgoDex, whereas EgoXtreme is evaluated using its complete available release because its total duration is substantially smaller. The analysis therefore focuses not only on the breadth of the visual and semantic distributions, but also on the completeness, consistency, and practical usability of the associated supervision.

5.1. High-Dimensional Visual Diversity

We use CLIP image embeddings to compare the visual feature distributions of the four datasets. Visual frames are sampled from each video sequence and encoded using the CLIP image encoder. Each embedding is then ℓ_2 -normalized, yielding $\mathbf{z}_i \in \mathbb{R}^D$, where i indexes a sampled visual observation and D is the CLIP feature dimension. To control for unequal dataset sizes, we conduct 3,000 balanced random trials, each sampling 2,000 embeddings without replacement from every dataset. All reported values are averaged across these trials.

We first examine the covariance spectrum of each dataset in the original normalized CLIP space. For dataset s , let $\lambda_{s,j}$ denote the j -th eigenvalue of its embedding covariance matrix, where $j = 1, \dots, D$, and let $p_{s,j}$ be the corresponding proportion of total variance. Effective Rank is the exponential entropy of the normalized covariance spectrum, whereas Participation Ratio

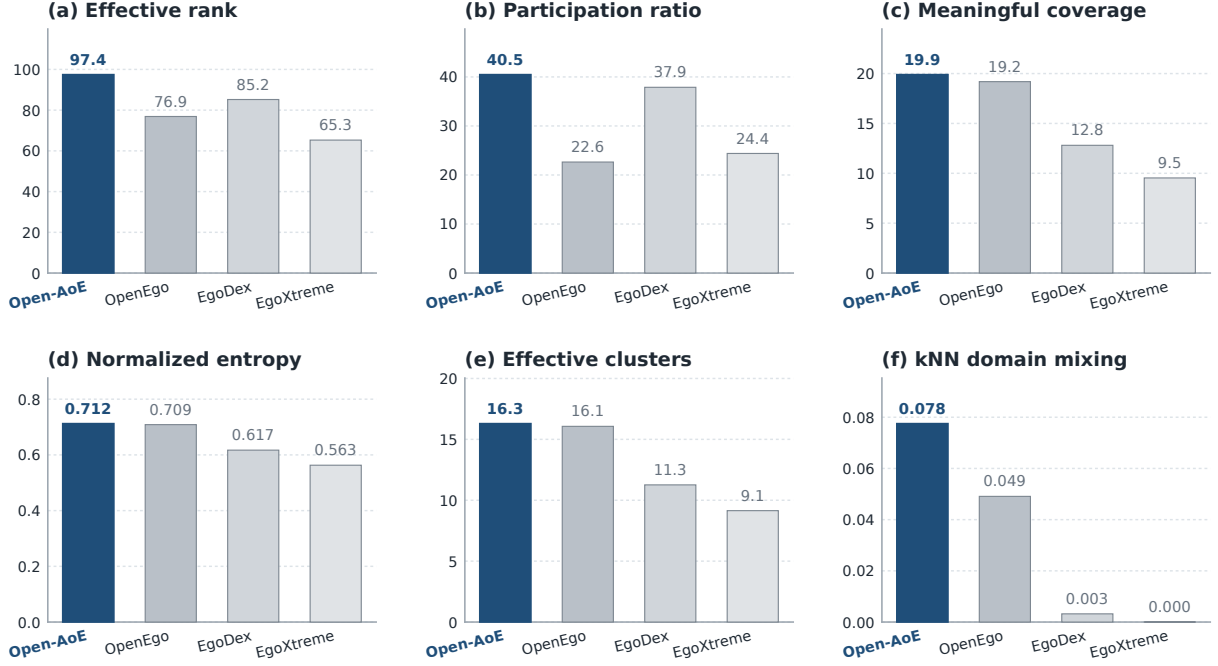


Figure 6: High-dimensional CLIP feature analysis. Bars report the mean over 3,000 balanced random subsamples, with 2,000 embeddings drawn from each dataset in every trial. Open-AoE has the highest mean in all six metrics.

measures its inverse concentration:

$$p_{s,j} = \frac{\lambda_{s,j}}{\sum_r \lambda_{s,r}}, \quad R_{\text{eff}}^{(s)} = \exp\left(-\sum_j p_{s,j} \log p_{s,j}\right), \quad R_{\text{PR}}^{(s)} = \frac{\left(\sum_j \lambda_{s,j}\right)^2}{\sum_j \lambda_{s,j}^2}. \quad (1)$$

Both metrics estimate the effective number of feature directions that carry substantial variance. Higher values therefore indicate that visual variation is distributed across a larger set of active directions rather than being concentrated in a few dominant modes.

We next measure how broadly and evenly each dataset occupies a shared visual feature partition. In each trial, the pooled balanced embeddings are transformed using a common PCA representation and partitioned into a shared codebook of $K = 50$ cells. Let $n_{s,c}$ denote the number of embeddings from dataset s assigned to cell c , and let $q_{s,c}$ denote the corresponding occupancy proportion. Meaningful Coverage counts cells containing at least τ samples, while Normalized Cluster Entropy and Effective Clusters quantify the balance of the occupancy distribution:

$$q_{s,c} = \frac{n_{s,c}}{\sum_{u=1}^K n_{s,u}}, \quad C_{s,\tau} = \sum_{c=1}^K \mathbf{1}[n_{s,c} \geq \tau], \quad (2)$$

$$H_{\text{norm}}^{(s)} = -\frac{\sum_{c=1}^K q_{s,c} \log q_{s,c}}{\log K}, \quad N_{\text{eff}}^{(s)} = \exp\left(-\sum_{c=1}^K q_{s,c} \log q_{s,c}\right).$$

Here, $C_{s,\tau}$ denotes the number of visual regions with sufficiently stable sample support, with $\tau = 5$ used throughout our experiments. $H_{\text{norm}}^{(s)}$ measures how evenly the samples are distributed across the shared codebook, while $N_{\text{eff}}^{(s)}$ expresses the same entropy as an equivalent number of uniformly occupied cells. Higher values indicate broader and more balanced coverage of the shared visual feature space.

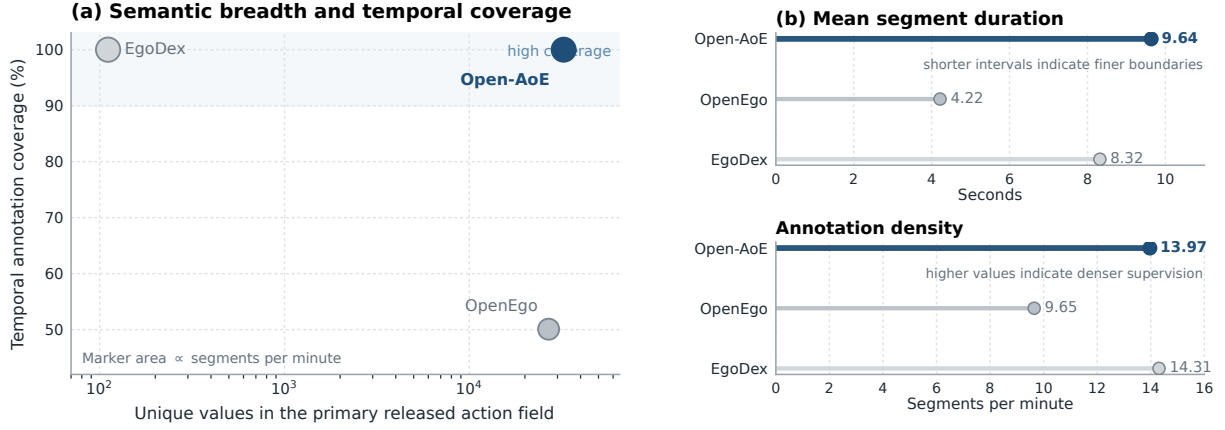


Figure 7: **Semantic breadth and temporal annotation characteristics across datasets.** (a) Number of unique entries in the primary released action-annotation field versus temporal annotation coverage, with marker area proportional to annotation density, measured as the number of annotated segments per minute. (b) Mean segment duration and annotation density for each dataset. Because the datasets use different native annotation schemas, vocabulary size reflects the breadth of the released semantic supervision rather than a controlled vocabulary comparison under a shared ontology.

Finally, we compute kNN Domain Mixing directly in the original normalized CLIP space, without codebook discretization. Let y_i denote the dataset identity of embedding i , let N_s be the number of sampled embeddings from dataset s , and let $\mathcal{N}_k(i)$ denote the set of its k nearest neighbors in the pooled sample. The metric is defined as

$$M_{\text{kNN}}^{(s)} = 1 - \frac{1}{N_s} \sum_{i: y_i = s} \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} \mathbf{1}[y_j = s], \quad (3)$$

where $\mathbf{1}[\cdot]$ is the indicator function and $k = 20$. The inner term measures the proportion of same-dataset neighbors around an embedding; therefore, $M_{\text{kNN}}^{(s)}$ is the average proportion of neighboring samples originating from other datasets. A higher value indicates stronger local connectivity to a broader range of visual domains.

Across the 3,000 balanced trials, Open-AoE achieves the highest mean in all six metrics: Effective Rank (97.43), Participation Ratio (40.47), Meaningful Coverage (19.89), Normalized Cluster Entropy (0.712), Effective Clusters (16.29), and kNN Domain Mixing (0.0775). The largest advantages are observed in Effective Rank and kNN Domain Mixing, indicating that Open-AoE spans a larger number of active feature directions while maintaining stronger local connectivity to other visual domains. Moreover, Open-AoE ranks first in Effective Rank, Participation Ratio, and kNN Domain Mixing in all 3,000 trials. Its advantages in the coverage- and entropy-based metrics are smaller, particularly relative to OpenEgo, but their repeated-sampling means remain the highest among all four datasets.

5.2. Semantic Breadth and Temporal Completeness

We next examine the semantic breadth and temporal structure of the released action annotations. Temporal coverage is defined as the fraction of the evaluated sequence timeline covered by at least one action segment, measuring the continuity of semantic supervision over the underlying video. The released action vocabulary is quantified by the number of unique entries in each

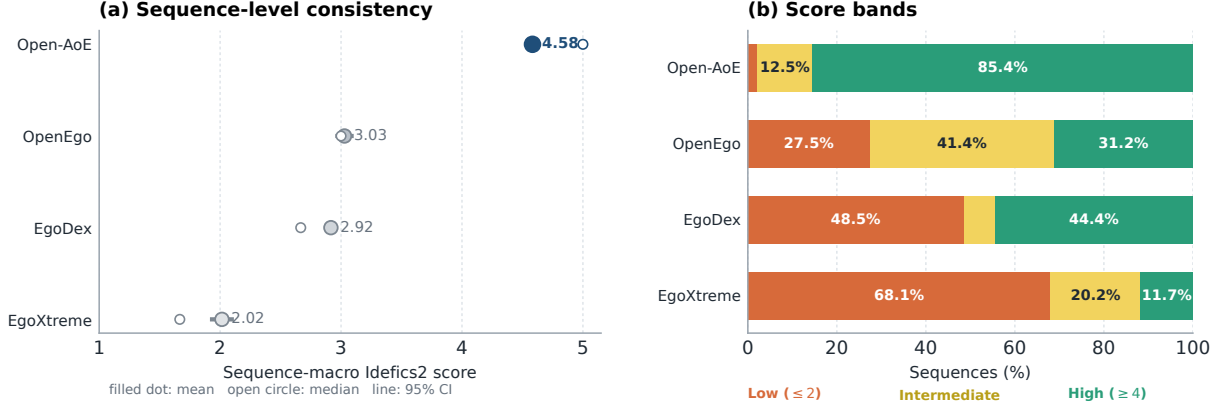


Figure 8: **Sequence-level image-annotation consistency across datasets.** (a) Sequence-macro means, sequence-level medians, and 95% bootstrap confidence intervals for the mean Idefics2 consistency score. (b) Fractions of sequences in the low (≤ 2), intermediate ($2 < \cdot < 4$), and high (≥ 4) score ranges.

dataset’s primary action-annotation field. Because the datasets follow different native annotation schemas, this measure reflects the breadth of semantic supervision exposed by each release rather than a controlled comparison under a shared action ontology. We further characterize annotation structure using mean segment duration and annotation density, defined as the number of annotated segments per minute. The former captures the typical temporal extent of individual action annotations, whereas the latter reflects their frequency along the video timeline. Taken together, these metrics capture complementary aspects of supervision: vocabulary size reflects semantic breadth, temporal coverage measures annotation completeness, and segment duration and density characterize the granularity of localized action annotation.

As illustrated in Fig. 7, Open-AoE combines broad natural-language supervision with near-complete temporal coverage. Its primary action field comprises 32,407 distinct natural-language descriptions, compared with 26,864 native labels in OpenEgo and 111 categorical action classes in EgoDex. Beyond these descriptions, Open-AoE includes structured semantic fields with 8,030 object strings, 175 action verbs, and 135 scene labels, enabling each action interval to be represented at multiple levels of semantic abstraction. Its annotations cover 99.99% of the evaluated timeline, substantially exceeding the 50.1% coverage of OpenEgo, while retaining a high annotation density of 13.97 segments per minute. EgoDex achieves complete clip-level label coverage and a slightly higher segment density, but relies on a substantially smaller categorical action vocabulary. Overall, Open-AoE combines open-vocabulary semantic breadth with near-continuous temporal alignment and densely localized action supervision, without compromising annotation completeness.

5.3. Image-Annotation Consistency

We assess whether the released action annotations are supported by the corresponding visual content using Idefics2 as a shared vision-language evaluator. Let $a_{q,t} \in [1, 5]$ denote the consistency score assigned to the t -th sampled frame of sequence q . The sequence-macro score is defined as

$$S_{\text{macro}} = \frac{1}{Q} \sum_{q=1}^Q \left(\frac{1}{T_q} \sum_{t=1}^{T_q} a_{q,t} \right), \quad (4)$$

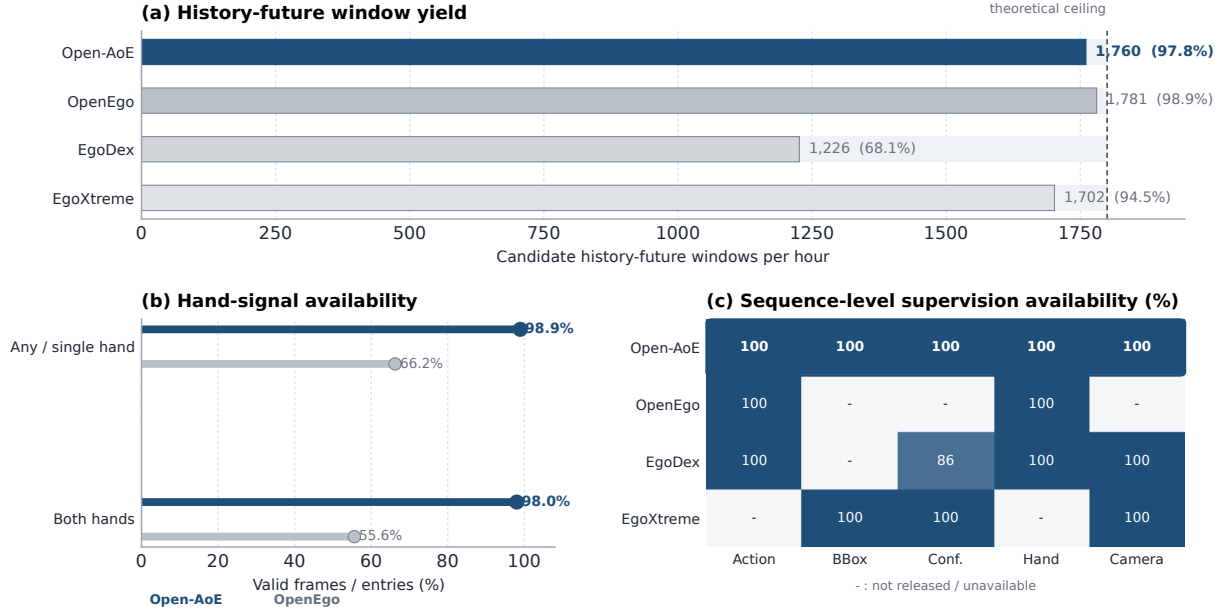


Figure 9: **Training-sample yield and multimodal supervision availability across datasets.** (a) Candidate history-future windows obtained per hour using 2 s of history, 2 s of future context, and a 2 s stride, together with the retention rate relative to the no-boundary ceiling of 1,800 windows per hour. (b) Availability of native hand signals, distinguishing entries with at least one valid hand from those with valid signals for both hands. (c) Sequence-level availability of action annotations, bounding boxes, annotation confidence, hand pose, and camera pose; “-” indicates that the corresponding field is not released or unavailable.

where Q is the number of evaluated sequences and T_q is the number of sampled frames in sequence q . Frame-level scores are first averaged within each sequence and then across sequences, thereby assigning equal weight to each sequence and preventing longer recordings from dominating the dataset-level estimate. Higher scores indicate stronger agreement between the released annotations and the sampled visual evidence.

As shown in Fig. 8, Open-AoE achieves a sequence-macro score of 4.583/5, substantially exceeding those of OpenEgo (3.029), EgoDex (2.916), and EgoXtreme (2.015). Its 95% bootstrap confidence interval is 4.557-4.608. This advantage is also evident in the score distribution: 85.4% of Open-AoE sequences attain a mean consistency score of at least 4, whereas only 2.2% score at or below 2. The higher aggregate score therefore reflects a broad concentration of sequences near the upper end of the scoring range rather than a small subset of exceptionally high-scoring cases. Because the datasets differ in their native annotation schemas and levels of textual detail, these results should be interpreted as a cross-dataset audit of image-annotation support under a shared evaluator rather than as a fully controlled benchmark of annotation accuracy.

5.4. Training-Window Retention and Multimodal Completeness

We finally assess how efficiently the released sequences can be converted into temporally aligned training samples and whether the principal supervision signals required by downstream embodied-learning pipelines are jointly available. For a sequence q of duration L_q , we construct candidate samples using a history horizon of $h = 2$ s, a future horizon of $f = 2$ s, and a temporal

stride of $\Delta = 2$ s. The number of candidate windows is defined as

$$N_q = \max\left(0, \left\lfloor \frac{L_q - h - f}{\Delta} \right\rfloor + 1\right). \quad (5)$$

The history-future window yield and normalized retention are then defined as

$$Y_{\text{window}} = \frac{\sum_q N_q}{\sum_q L_q / 3600}, \quad \eta_{\text{window}} = \frac{Y_{\text{window}}}{3600 / \Delta}, \quad (6)$$

where η_{window} measures retention relative to the no-boundary ceiling of $3600 / \Delta = 1,800$ windows per hour. This metric isolates losses induced by sequence boundaries and temporal discontinuities, excluding task-specific filtering imposed by individual downstream models. We additionally evaluate native hand-signal availability using the released validity information, distinguishing between entries with at least one valid hand and those with valid signals for both hands. At the sequence level, we record the availability of five principal supervision modalities: action annotations, bounding boxes, annotation confidence, hand pose, and camera pose.

As shown in Fig. 9, Open-AoE yields approximately 1,760 candidate history-future windows per hour, corresponding to 97.8% of the theoretical ceiling. This small gap from the no-boundary limit indicates that its temporal continuity enables dense extraction of fixed-context training samples with minimal loss at sequence boundaries. Open-AoE also exhibits substantially higher native hand-signal availability than OpenEgo, with 98.93% of evaluated entries containing at least one valid hand signal and 98.02% containing valid signals for both hands. Moreover, it is the only evaluated release in which all five supervision modalities are available with near-universal sequence-level coverage. Bounding-box quality is similarly high, with 100% valid boxes, 98.39% fully contained within the image boundaries, a mean confidence of 0.947, and a 10th-percentile confidence of 0.950. These results show that Open-AoE combines high temporal sample retention with dense, jointly available semantic, geometric, hand-motion, and camera-motion supervision, thereby reducing the modality filtering and temporal alignment required before downstream VLA, world-model, and human-to-robot training.

6. Conclusion

We presented Open-AoE, an open data infrastructure that couples approximately 2,000 hours of real-world egocentric manipulation data captured with consumer smartphones with a complete path from data to models. On the production side, the capture and processing pipeline turns raw smartphone video into structured samples aligned across vision, language, hand motion, camera trajectories, and action boundaries; on the consumption side, the open-source toolchain supports synchronized visualization, 4D reconstruction, cross-embodiment retargeting, and training representations for Vision-Language-Action policies, World Action Models, and World Models. Its broad coverage of scenes, actions, interaction objects, participants, device models, and fields of view allows this infrastructure to connect low-cost real-world collection with multiple embodied-learning paradigms.

Open-AoE is intended not as a static data package, but as an open ecosystem co-developed by data contributors, universities, research institutions, robot developers, and model teams. We invite the community to contribute new real-world scenes and tasks, reconstruction and retargeting methods, training adapters, and evaluation results, allowing data, tools, and models to evolve together in an open loop. Our goal is to make Open-AoE the lowest-barrier egocentric data infrastructure for embodied intelligence—making egocentric data easy for anyone to collect and use, so that data is no longer the bottleneck for embodied foundation models.

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